MCQ A · conditioning and independence

After S37.

For each question, only one option is correct. Justify your choice or checking the solutions.

1. $P_A(B)$ denotes:

A. $P(A \cap B)$ B. $P(B \mid A)$

C. $P(A \mid B)$ D. $P(\overline{B})$

2. If $P(A) = 0.4$ and $P_A(B) = 0.5$, then $P(A \cap B)$ is:

A. 0,9 B. 0,2

C. 0,5 D. 0,1

3. Independent events satisfy:

A. $P(A \cap B) = 0$ B. $P(A \cap B) = P(A) + P(B)$

C. $P(A \cap B) = P(A)P(B)$ D. $A = B$

4. "At most 3 successes" translates as:

A. $X < 3$ B. $X \leq 3$

C. $X \geq 3$ D. $X > 3$

5. In a weighted tree, the probability of a path is:

A. the sum of its branches B. the product of its branches

C. always $\frac{1}{2}$ D. a difference

MCQ B · binomial distribution and threshold

After S41; integer counting variables.

For each question, only one option is correct. Justify your choice orally before checking the solutions.

6. $X \sim \mathcal{B}(10; 0{,}3)$ counts:

A. the successes of ten identical and independent trials B. ten probabilities

C. the failures only D. a continuous mean

7. For $X \sim \mathcal{B}(n; p)$, $E(X)$ is:

A. p^n B. np

C. $n + p$ D. $n(1 - p)$

8. "Strictly more than 4" is equivalent to:

A. $X \geq 4$ B. $X \leq 4$

C. $X \geq 5$ D. $X > 5$

9. The assumptions of the Bernoulli scheme are not all satisfied if:

A. there is a fixed number of trials B. success has the same probability

C. the trials are dependent D. the outcome is success/failure

10. To check a minimal integer threshold in a monotonic problem, we also check:

A. the previous index B. a colour in the tree

C. $P(X = 0)$ only D. the name of X

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. $P_A(B) = P_B(A)$ in any situation.

2. If $X \sim \mathcal{B}(n; p)$, then $P(X \geq k) = 1 - P(X \leq k - 1)$.

3. An observed frequency is exactly equal to the theoretical probability.

Review of prior work

I know how to build a tree, choose or refuse the binomial model and translate an integer threshold. **To watch out for:** swapping conditioning and intersection; forgetting to shift an integer bound.